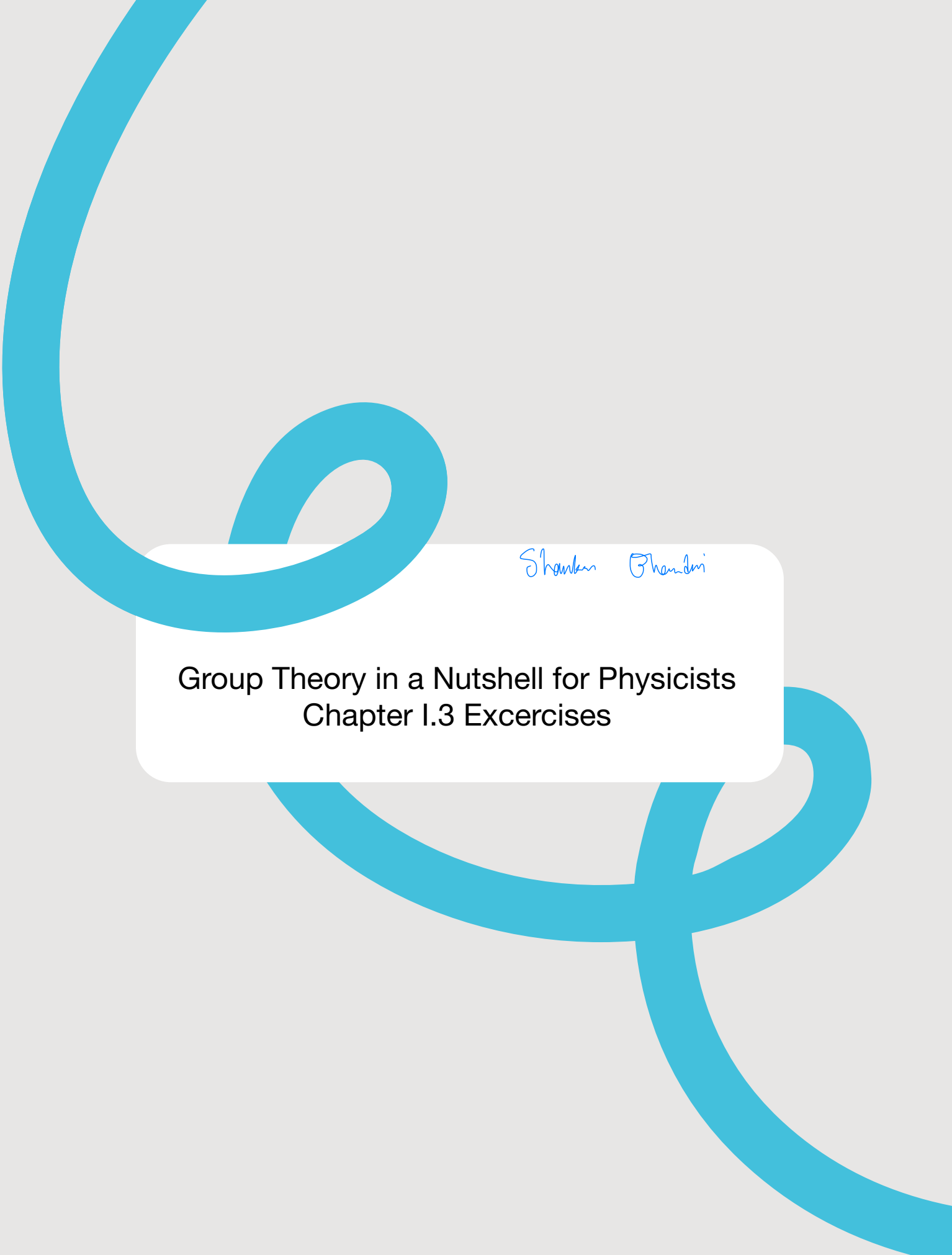




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Group Theory in a Nutshell for Physicists
Chapter I.3 Exercises

- 1 Suppose you are given two vectors $\vec{p}$ and $\vec{q}$ in ordinary 3-dimensional space. Consider this array of three numbers:

$$\begin{pmatrix} p^2 q^3 \\ p^3 q^1 \\ p^1 q^2 \end{pmatrix}$$

Prove that it is not a vector, even though it looks like a vector. (Check how it transforms under rotation!) In contrast,

$$\begin{pmatrix} p^2 q^3 - p^3 q^2 \\ p^3 q^1 - p^1 q^3 \\ p^1 q^2 - p^2 q^1 \end{pmatrix}$$

does transform like a vector. It is in fact the vector cross product $\vec{p} \otimes \vec{q}$.

$$\vec{p} = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}, \quad \vec{q} = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}$$

$$U = \begin{pmatrix} p^2 q^3 \\ p^3 q^1 \\ p^1 q^2 \end{pmatrix}, \quad R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

in 2D rotation

Under rotation we have

$$\vec{p}' = R(\theta) \vec{p} \Rightarrow p_1 \rightarrow p'_1 \text{ and } p_2 \rightarrow p'_2$$

$$\vec{q}' = R(\theta) \vec{q} \Rightarrow q_1 \rightarrow q'_1 \text{ and } q_2 \rightarrow q'_2$$

$$\text{With: } \vec{p}' = R(\theta) \vec{p} = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} = \begin{pmatrix} \cos \theta p_1 + \sin \theta p_2 \\ -\sin \theta p_1 + \cos \theta p_2 \\ p_3 \end{pmatrix} = \begin{pmatrix} p'_1 \\ p'_2 \\ p'_3 \end{pmatrix}$$

$$\text{Similarly } \vec{q}' = R(\theta) \vec{q} = \begin{pmatrix} \cos \theta q_1 + \sin \theta q_2 \\ -\sin \theta q_1 + \cos \theta q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} q'_1 \\ q'_2 \\ q'_3 \end{pmatrix}$$

$R(\theta) U$ is

$$R(\theta) U = \begin{pmatrix} \cos \theta p_2 q_3 + \sin \theta p_3 q_1 \\ -\sin \theta p_1 q_3 + \cos \theta p_3 q_1 \\ p_1 q_2 \end{pmatrix}$$

but from plugging $\vec{p}'$ & $\vec{q}'$ that are real vectors we get

$$U' = \begin{pmatrix} p'_2 \cdot q'_3 \\ p'_3 \cdot q'_1 \\ p'_1 \cdot q'_2 \end{pmatrix} = \begin{pmatrix} (-\sin \theta p_1 + \cos \theta p_2) q_3 \\ p_3 (\cos \theta q_1 + \sin \theta q_2) \\ (\cos \theta p_1 + \sin \theta p_2) (-\sin \theta q_1 + \cos \theta q_2) \end{pmatrix} \neq R(\theta) U$$

$\Rightarrow V$ doesn't transform as a vector $\Rightarrow V$ is not a vector

Now let's check up on

$$V = \begin{pmatrix} p^2 q^3 - p^3 q^2 \\ p^3 q^1 - p^1 q^3 \\ p^1 q^2 - p^2 q^1 \end{pmatrix}$$

Under rotation we have:

$$R(\theta)V = \begin{bmatrix} \cos\theta (p_2 q_3 - p_3 q_2) + \sin\theta (p_3 q_1 - p_1 q_3) \\ -\sin\theta (p_2 q_3 - p_3 q_2) + \cos\theta (p_3 q_1 - p_1 q_3) \\ p^1 q^2 - p^2 q^1 \end{bmatrix}$$

And via $\vec{p}'$ & $\vec{q}'$ we get V'

$$V' = \begin{pmatrix} p_2' q_3' - p_3' q_2' \\ p_3' q_1' - p_1' q_3' \\ p_1' q_2' - p_2' q_1' \end{pmatrix} \quad \left\{ \begin{array}{l} p_3' = p_3, \quad q_3' = q_3 \\ p_1' = \cos\theta p_1 + \sin\theta p_2 \\ p_2' = -\sin\theta p_1 + \cos\theta p_2 \end{array} \right.$$

$$= \begin{pmatrix} q_3 (-\sin\theta p_1 + \cos\theta p_2) - p_3 (-\sin\theta q_1 + \cos\theta q_2) \\ p_3 (\cos\theta q_1 + \sin\theta q_2) - q_3 (\cos\theta p_1 + \sin\theta p_2) \\ (\cos\theta p_1 + \sin\theta p_2)(-\sin\theta q_1 + \cos\theta q_2) - (-\sin\theta p_1 + \cos\theta p_2)(\cos\theta q_1 + \sin\theta q_2) \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta (p_2 q_3 - p_3 q_2) + \sin\theta (p_3 q_1 - p_1 q_3) \\ -\sin\theta (p_2 q_3 - p_3 q_2) + \cos\theta (p_3 q_1 - p_1 q_3) \\ p_1 q_2 [\cos^2\theta + \sin^2\theta] + p_1 q_1 [-\cos\theta \sin\theta + \cos\theta \sin\theta] + p_2 q_1 [-\sin^2\theta - \cos^2\theta] + p_2 q_2 [\sin\theta \cos\theta - \sin\theta \cos\theta] \end{pmatrix}$$

$$\Rightarrow V' = \begin{pmatrix} \cos\theta (P_1 q_3 - P_3 q_1) + \sin\theta (P_3 q_1 - P_1 q_3) \\ -\sin\theta (P_2 q_3 - P_3 q_2) + \cos\theta (P_3 q_1 - P_1 q_3) \\ P_1 q_2 - P_2 q_1 \end{pmatrix} = R(\theta) V$$

$\Rightarrow V$ is a vector

2. Verify that $R \approx I + A$, with A given by $A = \theta_x I_x + \theta_y I_y + \theta_z I_z$, satisfies the condition $\det R = 1$

$$\Rightarrow \det(R) = \det(I + A)$$

$$I + A = I + \theta_x \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} + \theta_y \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} + \theta_z \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow I + A = \begin{pmatrix} 1 & \theta_z & -\theta_y \\ -\theta_z & 1 & \theta_x \\ \theta_y & -\theta_x & 1 \end{pmatrix}$$

$$\Rightarrow \det(I + A) = 1 \cdot \begin{vmatrix} 1 & \theta_x \\ -\theta_x & 1 \end{vmatrix} - \theta_z \underbrace{\begin{vmatrix} -\theta_z & \theta_x \\ \theta_y & 1 \end{vmatrix}}_{\substack{\text{of order } \theta_z^2 \\ \rightarrow 0}} - \theta_y \underbrace{\begin{vmatrix} -\theta_z & 1 \\ \theta_y & -\theta_x \end{vmatrix}}_{\substack{\text{of order } \theta_y^2 \\ \rightarrow 0}}$$

$= 1$ to order of θ_x $\underbrace{\hspace{10em}}_{\text{since } \theta_x, \theta_y, \theta_z \text{ are infinitesimal}}$

$$\Rightarrow \det(I + A) = 1(1 + \theta_x^2)$$

$$= 1 \quad \text{to order of } \theta_x$$

$$\Rightarrow \det(R) = \det(I + A) = 1$$

$$\Rightarrow \det(R) = 1 \quad !!!$$

3. Using eq. (14), show that a rotation around the x axis through angle θ_x is given by

$$R_x(\theta_x) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & R_2(\theta) \end{pmatrix} \quad \text{where} \quad R_2(\theta) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

The generator of rotation around x axis is

$$J_x$$

$$\Rightarrow R_x(\theta_x) = e^{\theta_x J_x} \quad \text{and} \quad J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$i^2 = J_x^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = - \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{I} \end{pmatrix} = -1$$

$$J_x^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = -J_x \quad -i$$

$$\Rightarrow J_x^4 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow J_x^5 = J_x \quad (-i)^4 = 1$$

Our J_x behaves like i in the sense that the 2x2 matrix of bottom right corner (denoted K) has $K^2 = -1$. As such we can write

$$e^{\theta_x J_x} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix} + \cos\theta \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} + \sin\theta \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix} \quad \text{Hm,}$$

Let's again in next page more rigorously:

$$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$J^0 = I$$

$$e^{\theta_x J_x} = \sum_{n=0}^{\infty} \frac{\theta_x^n}{n!} J_x^n$$

$$= \sum_{k=0}^{\infty} \theta^{2k} \frac{J_x^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \theta^{2k+1} \frac{J_x^{2k+1}}{(2k+1)!}$$

$$J^{2k} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -I_2 \end{pmatrix} = (-1)^k \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix}}_{\text{Constant matrix}} \quad \text{except for } k=0 \text{ when } J^0 = I_3$$

$$J^{2k+1} = (-1)^k \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} J = (-1)^k \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$J^{2k+1} = (-1)^k J$$

$$\Rightarrow \exp(\theta_x J_x) = \left(\sum_{k=1}^{\infty} (-1)^k \frac{\theta_x^{2k}}{(2k)!} \right) \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} + I_3 + \left(\sum_{k=0}^{\infty} (-1)^k \frac{\theta_x^{2k+1}}{(2k+1)!} \right) J$$

We recognise the second sum as $\sin(\theta_x)$

We can also write

$$\begin{aligned} \left(\sum_{k=1}^{\infty} (-1)^k \frac{\theta_x^{2k}}{(2k)!} \right) \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} &= \left(\sum_{k=0}^{\infty} (-1)^k \frac{\theta_x^{2k}}{(2k)!} \right) \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} \cos \theta_x - \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \Rightarrow \exp(\theta_x J_x) &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} \cos \theta_x + I_3 - \begin{pmatrix} 0 & 0 & 0 \\ 0 & I_2 \end{pmatrix} + \sin \theta J_x \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & \sin \theta_x \\ 0 & -\sin \theta_x & \cos \theta_x \end{pmatrix} \quad \square \end{aligned}$$

Also write down $R_y(\theta_y)$ & show $R_x(\theta_x) R_y(\theta_y) \neq R_y R_x$

$$R_y(\theta_y) = \exp(\theta_y J_y) \quad \text{with} \quad J_y = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$= \sum_{n=0}^{\infty} \theta^n \frac{J_y^n}{n!}$$

$$J_y^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \sum_{k=0}^{\infty} \theta^{2k} \frac{J_y^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \theta^{2k+1} \frac{J_y^{2k+1}}{(2k+1)!}$$

$$J_y^{2k} = (-1)^k \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{with} \quad J_y^0 = I$$

$$\text{and} \quad J_y^{2k+1} = (-1)^k J_y$$

$$\Rightarrow R_y(\theta_y) = \left(\sum_{k=1}^{\infty} \theta^{2k} \frac{1}{(2k)!} (-1)^k \right) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + I_3 + \left(\sum_{k=0}^{\infty} (-1)^k \theta^{2k+1} \frac{1}{(2k+1)!} \right) J_y$$

$$= \cos \theta_y \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + I_3 - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sin \theta_y J_y$$

$$R_y(\theta_y) = \begin{pmatrix} \cos \theta_y & 0 & -\sin \theta_y \\ 0 & 1 & 0 \\ \sin \theta_y & 0 & \cos \theta_y \end{pmatrix}$$

$$\Rightarrow R_y(\theta_y) R_x(\theta_x) = \begin{pmatrix} \cos \theta_y & 0 & -\sin \theta_y \\ 0 & 1 & 0 \\ \sin \theta_y & 0 & \cos \theta_y \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & \sin \theta_x \\ 0 & -\sin \theta_x & \cos \theta_x \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta_y & \sin \theta_x \sin \theta_y & -\sin \theta_y \cos \theta_x \\ 0 & \cos \theta_x & \sin \theta_x \\ \sin \theta_y & -\cos \theta_y \sin \theta_x & \cos \theta_y \cos \theta_x \end{pmatrix}$$

$$R_x(\theta_x) R_y(\theta_y) = \begin{pmatrix} \cos \theta_y & 0 & -\sin \theta_y \\ \vdots & \ddots & \vdots \\ \vdots & \ddots & \vdots \end{pmatrix}$$

$\Rightarrow$ just from top row we see that

$$R_y R_x \neq R_x R_y$$

4 Use the Hermiticity of J to show C_{ijk} in eq (18) are real numbers.

eq. (18): $[J_i, J_j] = i C_{ijk} J_k$

Now we do a funny thing of taking adjoint of both sides

$$[J_i, J_j]^\dagger = (i C_{ijk} J_k)^\dagger \quad \parallel \quad \text{Here we assume } C_{ijk} \text{ is some complex number not necessarily real}$$

$$\Rightarrow [J_i^\dagger, J_j^\dagger] = -i C_{ijk}^\dagger J_k^\dagger \quad \parallel \quad [J_i, J_j]^\dagger = (J_i J_j - J_j J_i)^\dagger$$

$$\Rightarrow = J_i^\dagger J_j^\dagger - J_j^\dagger J_i^\dagger$$

Hermiticity of J_i & J_j tells us that

$$J_i^\dagger = J_i$$

$$\Rightarrow [J_j, J_i] = -i C_{ijk}^\dagger J_k^\dagger \quad \parallel \quad J_k^\dagger = J_k \quad \& \quad [J_j, J_i] = -[J_i, J_j]$$

$$[J_i, J_j] = i C_{ijk}^\dagger J_k$$

Since we also started from $[J_i, J_j] = i C_{ijk} J_k$

$$\Rightarrow C_{ijk}^\dagger = C_{ijk}^* = C_{ijk} \quad \parallel \quad C_{ijk}^\dagger = C_{ijk}^* \text{ since it's a complex number}$$

$$\Rightarrow C_{ijk} = C_{ijk}^* \text{ means that } \text{to don't have transpose.}$$

C_{ijk} is real.

5. Calculate $[J_{mn}, J_{pq}]$ by brute force using eq. (24)

$$\text{Eq. (24): } (J_{mn})^{ij} = -i (\delta^{mi} \delta^{nj} - \delta^{mj} \delta^{ni})$$

$$\Rightarrow [J_{mn}, J_{pq}]^{ij} = (J_{mn} J_{pq} - J_{pq} J_{mn})^{ij} \quad \left\{ \begin{array}{l} (AB)^{ik} = \sum_j A^{ij} B^{jk} \\ (AB)^{ik} = A^{ij} B^{jk} \\ \text{we assume} \\ \text{repeat indices sum} \end{array} \right.$$

$$(J_{mn} J_{pq})^{ij} = J_{mn}^{ik} J_{pq}^{kj}$$

$$= -i (\delta^{mi} \delta^{nk} - \delta^{mk} \delta^{ni}) (-i) (\delta^{pk} \delta^{qj} - \delta^{pj} \delta^{qk})$$

$$= - (\delta^{mi} \delta^{nk} \delta^{pk} \delta^{qj} - \delta^{mi} \delta^{nk} \delta^{pj} \delta^{qk} - \delta^{mk} \delta^{ni} (\delta^{pk} \delta^{qj} - \delta^{pj} \delta^{qk}))$$

$$= - (\delta^{np} \delta^{mi} \delta^{qj} - \delta^{nq} \delta^{mi} \delta^{pj} - \delta^{mp} \delta^{ni} \delta^{qj} + \delta^{mq} \delta^{ni} \delta^{pj})$$

$$= i^2 \delta^{mi} (\delta^{np} \delta^{jq} - \delta^{nq} \delta^{jp}) + i^2 \delta^{ni} (\delta^{mq} \delta^{jp} - \delta^{mp} \delta^{jq}) \quad \left\| \begin{array}{l} \text{use eq.} \\ (24) \end{array} \right.$$

$$= -i \delta^{mi} J_{(nq)}^{pq} - i \delta^{ni} J_{(mj)}^{qp}$$

$$[J_{mn}, J_{pq}]^{ij} = (J_{mn} J_{pq})^{ij} - (J_{pq} J_{mn})^{ij}$$

$$= \delta^{mi} (\delta^{nq} \delta^{jp} - \delta^{np} \delta^{jq}) + \delta^{ni} (\delta^{mp} \delta^{jq} - \delta^{mq} \delta^{jp}) \\ - [\delta^{pi} (\delta^{qn} \delta^{jm} - \delta^{qm} \delta^{jn}) + \delta^{qi} (\delta^{pn} \delta^{jm} - \delta^{pn} \delta^{jm})]$$

$$= \delta^{nq} [\delta^{mi} \delta^{jp} - \delta^{jm} \delta^{pi}] - \delta^{np} [\delta^{mi} \delta^{jq} - \delta^{qi} \delta^{jm}] \\ + \delta^{mp} [\delta^{ni} \delta^{jq} - \delta^{qi} \delta^{jn}] - \delta^{mq} [\delta^{ni} \delta^{pj} - \delta^{pi} \delta^{jn}]$$

$$= \delta_{nq} [\delta^{mi} \delta^{pj} - \delta^{mj} \delta^{pi}] + \delta_{mp} [\delta^{ni} \delta^{qj} - \delta^{nj} \delta^{qi}] \\ - \delta_{np} [\delta^{mi} \delta^{qj} - \delta^{mj} \delta^{qi}] - \delta_{mq} [\delta^{ni} \delta^{pj} - \delta^{nj} \delta^{pi}]$$

Now
use Eq. (24)
again

$$= i (\delta_{mp} J_{(nq)}^{ij} + \delta_{nq} J_{(mp)}^{ij} - \delta_{np} J_{(mq)}^{ij} - \delta_{mq} J_{(np)}^{ij})$$

$$\Rightarrow [J_{(mn)}, J_{(pq)}]^{ij} = i (\delta_{mp} J_{(nq)}^{ij} + \delta_{nq} J_{(mp)}^{ij} - \delta_{np} J_{(mq)}^{ij} - \delta_{mq} J_{(np)}^{ij})$$

$$\Rightarrow [J_{(mn)}, J_{(pq)}] = i (\delta_{mp} J_{(nq)} + \delta_{nq} J_{(mp)} - \delta_{np} J_{(mq)} - \delta_{mq} J_{(np)}) \quad \square$$

6. Of the six 4-by-4 matrices $J_{12}, J_{23}, J_{31}, J_{14}, J_{24}, J_{34}$ that generate $SO(4)$, what is the maximum number that can be simultaneously diagonalized?

The matrices A, B can be simultaneously diagonalized if

$$[A, B] = 0$$

For $SO(4)$ we have the following commutator relations:

$$[J_i, J_j] = i \varepsilon_{ijk} J_k \quad \text{with} \quad J_i = \{J_{23}, J_{31}, J_{12}\}$$

$$[J_i, K_j] = i \varepsilon_{ijk} K_k \quad K_i = \{J_{14}, J_{24}, J_{34}\}$$

$$[K_i, K_j] = i \varepsilon_{ijk} J_k$$

$$\Rightarrow [J_1, J_2] = i \varepsilon_{123} J_3 \quad \Rightarrow \text{None of the } J\text{'s commute with each other}$$

$$[J_1, J_3] = i \varepsilon_{132} J_2$$

$$[J_2, J_3] = i \varepsilon_{231} J_1$$

$$[J_i, K_i] = i \varepsilon_{iik} J_k = 0 \quad \text{so } J_i \text{ \& } K_j \text{ commute for } i=j$$

The only zero commutators are $[J_1, K_1] = [J_2, K_2] = [J_3, K_3] = 0$

Note tho that there 3 are distinct commutators so

J_1 & J_2 don't commute same with K , J_2 etc.

So the maximum number that can be simultaneously diagonalized is 2. Since the maximum number of matrices that mutually commute is 2 (although there are 3 such 2's).